

Basic Possibilistic Logic: Examples and Visualization

Seminar Paper

submitted by

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Abstract

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1 Introduction

The goal of this paper is to introduce Basic Possibilistic Logic with the use of examples and visualization. Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas. Its strength lies in modeling uncertainty and its ability to deal with inconsistency. It has been developed for about thirty years. It still remains very close to classical logic. In this paper we will only focus on basic Possibilistic Logic.

This work is structured as follows. In the Section 2 Basic Possibilistic Logic is explained with its Preliminaries for understanding the syntax and semantics of the logic. The Section 3 shows the use of Basic Possibilistic Logic with multiple examples. Section 4 explains Basic Possibilistic Logic with graphical explanations. The last Section 5 discusses the results of the previous sections.

2 Fundamentals of Basic Possibilistic Logic

As a main source the overview chapter: Possibilistic Logic - An Overview from the authors Dubois and Prade [1] is used.

2.1 Introduction to Possibility Theory

2.1.1 Possibility Distribution

A possibility distribution is a mapping π from U to a totally ordered scale S , with top denoted by 1 and bottom by 0:

$$\pi : U \rightarrow S,$$

where U is the set of states. $\pi(u)$ can be understood as the degree of possibility of the state u . The function π represents the state of knowledge of an agent and distinguishes what is plausible from what is less plausible. The state u is impossible, if $\pi(u)$ equals 0. The state is fully possible, if $\pi(u)$ equals 1.

A distribution π is at least as specific as π' if

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

An agent has complete knowledge if the possibility of one state is 1 and the possibility of all other states is 0. Complete ignorance is given if the possibility of all states is 1. The Principle of minimal specificity states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

2.1.2 Possibility and Necessity Functions

If we want to know if the event A occurs we can respond in possibility theory with two measures calculated from the possibility distribution to A . A is a subset of states. The two measures are the possibility and the necessity of an event.

The Possibility is defined as $\Pi(A) = \sup_{u \in A} \pi(u)$. It is the maximum of possibility of all the states of A . The Possibility evaluates the extent to which A is consistent with π . Possibility measures satisfy the characteristic maximality property $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$.

The Necessity is defined as $N(A) = \inf_{u \notin A} (1 - \pi(u))$. It is the minimum of the inversed possibility value of all states which are not in A . The Necessity evaluates the extent to which A is certainly implied by π . Necessity measures satisfy the characteristic minimization property $N(A \cap B) = \min(N(A), N(B))$. The minimization property expresses that to be certain about A and B , one has to be certain about A and to be certain about B .

The duality of the two measures is expressed as $N(A) = 1 - \Pi(A^c)$, where A^c is the complement of A .

2.2 Formal foundations of Basic Possibilistic Logic

2.2.1 Syntax

A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where a is a classical logic formula (e.g., $p \rightarrow q$) and α is the certainty level (minimum necessity degree of a). It can be viewed as a lower bound of a necessity measure. The pair (a, α) is understood as $N(a) > \alpha$. We can interpret that the formula a is at least certain to degree α . The higher α , the stronger the belief in a . For example $(b \rightarrow f, 0.9)$ means that birds fly with certainty 0.9.

2.2.2 Axioms

The axioms of Possibilistic Logic are those of propositional logic. In the following the inference rules of basic possibilistic logic will be explained. The Certainty weakening rule states that if a formula is valid with a high certainty level, it is also valid with any lower certainty level. If a formula holds with a certainty level α , it automatically implies that it holds with any lower level of certainty beta because we are being less demanding, see equation 1.

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta) \quad (1)$$

The Modus ponens rules states that if a to b is true with certainty level α , and a is also true with certainty level α then b must be true with certainty level α , see equation 2.

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1] \quad (2)$$

In classical logic, the resolution rule allows us to combine two clauses that share a literal to eliminate that literal and infer a new clause. In possibilistic logic, the resolution rule preserves the certainty level α , because the inferred formula $b \vee c$ cannot be more certain than the least certain premise, see equation 3.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha) \quad (3)$$

The weakest link resolution rule extends the resolution rule by the concept of different certainty levels α and β in the premises. The weakest link resolution reflects the idea that the certainty of the conclusion cannot exceed the weakest certainty of the premises. If one of the premises is less certain, it becomes the bottleneck for the certainty of the conclusion, see equation 4.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)) \quad (4)$$

2.2.3 Inference Rules

We can define a possibilistic knowledge base as a collection of possibilistic formulas, see equation 5.

$$\Gamma = \{(a_i, \alpha_i), i = 1, \dots, m\} \quad (5)$$

Further we can define the α -cut of knowledge base, see equation 6. It only contains those formulas from the origin knowledge base with a certainty level of at least α . It has a nested structure where the α knowledge base is a part of the β knowledge base if α is higher than β .

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\} \quad (6)$$

A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$, see equation 7.

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha) \quad (7)$$

If we want to infer from the knowledge, we can only look on the formulas with a certainty of at least α . This means that a inference is only valid if it is valid with the α -cut of a knowledge base.

An important attribute of knowledge bases is the inconsistency level. It is the highest certainty level at which the knowledge base becomes inconsistent. The properties of the inconsistency level shows that all formulas which have a higher certainty level than the inconsistency level are safe form inconsistency, see equation 8.

$$\text{inc-l}(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\perp, \alpha)\}. \quad (8)$$

2.2.4 Semantic Aspects

The semantics of possibilistic logic, as introduced in [?], is based on the concepts of possibility distributions, weak possibility measures, and strong necessity measures.

Consider a possibilistic logic formula (a, α) , representing the constraint $N(a) \geq \alpha$. The semantics of this formula is expressed using a possibility distribution $\pi(a, \alpha)$, defined as:

$$\pi(a, \alpha)(\omega) = \begin{cases} 1 & \text{if } \omega \models a, \\ 1 - \alpha & \text{if } \omega \models \neg a. \end{cases}$$

The intuitive interpretation is that models of a are fully possible ($\pi = 1$), while counter-models of a are less possible as α increases. When $\alpha = 0$, the formula $(a, 0)$ provides no additional information and can be ignored.

The associated necessity measure satisfies:

$$N_{a, \alpha}(a) = \alpha,$$

and $\pi(a, \alpha)$ is the least informative possibility distribution that satisfies this constraint.

2.3 Differences from classical logic

2.4 Relation to Artificial Intelligence

3 Analysis and Development of Examples

3.1 Elaboration on existing examples from the literature

The „Tweety penguin problem “

Other relevant examples from Dubois’ works

3.2 Extension of existing examples

Detailed discussion and interpretation

3.3 Development of new examples

Motivation for the new examples

Presentation and analysis

4 Visualization in Possibilistic Logic

4.1 The need for and benefits of visualization

4.2 Approaches to visualization

Graphical representations of possibility and necessity

Diagrams illustrating inference processes

4.3 Implementation and evaluation

5 Discussion and Conclusion

5.1 Summary of findings

5.2 Limitations of the study

5.3 Outlook on future research

6 Example Figures and Tables

Figure 1 shows an example of an abstract argumentation framework.

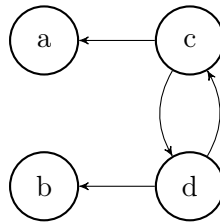


Figure 1: An abstract argumentation framework.

1	2	3
4	5	6

Table 1: Table caption.

6.1 Encoding Test

Städte, Länder, Flüsse

References

- [1] Didier Dubois and Henri Prade. Possibilistic logic — an overview. In Jörg H. Siekmann, editor, *Computational Logic*, volume 9 of *Handbook of the History of Logic*, pages 283–342. North-Holland, 2014.

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